

Blockchain-Based Transaction Validation and Management System

Yendamuri Nishanth, Tammalla Nikhil, Vadlamudi Sreevanth Chowdhary, Kavitha C. R.

Department of Computer Science and Engineering

Amrita School of Computing, Bengaluru

Amrita Vishwa Vidyapeetham, India

bl.en.u4cse22164@bl.students.amrita.edu,

bl.en.u4cse22156@bl.students.amrita.edu,

bl.en.u4cse22176@bl.students.amrita.edu,

cr_kavitha@blr.amrita.edu

Abstract—Our work represents a simplified version of a blockchain system utilizing Java, with a primary focus on transaction security, block creation, and blockchain management. It consists of the basic components of a blockchain: blocks, transaction pool, and a UTXO (Unspent Transaction Output) model to verify that all transactions are legitimate and correct. We also have applied multiple encryption methods in building our blockchain project and its main features: using the Vigenere cipher for basic symmetric encryption, utilizing AES (Advanced Encryption Standard) and DES (Data Encryption Standard) encryption to encrypt sensitive information at multiple stages of the transaction process, and using digital signatures to authenticate messages and prevent non-repudiation (having chain of trust in digital signatures helps verify identity logically and helps ensure that the transaction is not tampered). Every transaction must also be verified before it can be added to a block and ultimately a chain is extended based on consensus rules in a secure manner. This project can also be a useful educational resource for learning about facilitating basic blockchain knowledge, as well as the traditional cryptographic methods in a real-world scenario.

Index Terms—Blockchain, Transaction validation, Cryptography, Vigenere cipher, AES, DES, Digital signatures, UTXO, Java, Block creation, Chain validation, Consensus mechanism, Secure communication, Encryption techniques, Blockchain development.

I. INTRODUCTION

In today's rapidly changing digital world, the issue of verifying, safeguarding, and securing data exchanges is a critical concern. More users are accessing online systems for financial transactions, verification of documents, and conversing with other users than ever before. This has exposed the limitations of centralized systems in taking on the dichotomous challenges of cyber threats. Centralized systems are susceptible to data breaches, unauthorized use, or tampering, which can lead to serious consequences, such as identity theft, transactional fraud, and penetrating confidentiality. One possible approach to resolve such issues comes from blockchain technology, allowing for a decentralized, tamper-resistant, open, and auditable record. Blockchain differentiates itself from conventional systems because it disperses data among a network of nodes, and with this approach, it is practically impossible for rogue elements to make changes to the data without detection. But the viability of blockchain strongly depends on

implementing its cryptographic foundation blocks to ensure secure management of data.

Researchers have explored various options to improve the trustworthiness of blockchain by implementing cryptographic techniques. Blockchain timestamping have been used to provide long-term integrity for digital signatures, and access control using encryption schemes such as RSA or SHA-based hashing have also been used to secure access and verify identities. These approaches provide security against tampering, message authentication support, and enable non-repudiation. In spite of all these developments, most current implementations are based on standalone cryptographic approaches that do not address how various encryption schemes can be integrated and applied in practice within blockchain architectures. Therefore, the need remains for an all-encompassing, pedagogical blockchain platform that integrates both ancient and modern cryptographic techniques into a coherent, single framework.

Our effort addresses this lack by proposing a streamlined blockchain system coded in Java, meant to emphasize secure transaction verification, block generation, and overall integrity of the blockchain. The system proposes a multi-layered cryptography method to secure data during the course of transactions. Vigenere cipher is utilized for simple symmetric encryption of text data, providing an insight into traditional cryptographic operations. For high-security encryption, AES (Advanced Encryption Standard) and DES (Data Encryption Standard) are utilized for protecting sensitive transactional data at different points. Digital signatures are utilized to provide message authenticity, establish the identity of parties, and safeguard against unauthorized alteration of data thereby ensuring non-repudiation in the system.

One of the major characteristics of this blockchain is the use of the UTXO (Unspent Transaction Output) model that verifies the validity of transactions before adding them to a block. This prevents funds from being double-spent and ensures the accuracy and consistency of the ledger. Blocks are added based on consensus rules that serve to ensure chain integrity and eliminate conflicts. In addition, the work has involved the creation of an easy-to-use graphical interface for the user, enabling management of wallets, viewing public

keys, and securely transferring funds in the system's native cryptocurrency unit. The interface makes it easier to use and visually understand how blockchain operations actually function.

The contributions of our research are both technical advancement and educational accessibility. Not only does it show secure blockchain management and transaction verification, but it also provides hands-on experience with cryptographic concepts through practical application. By integrating multiple encryption algorithms with basic blockchain architecture, the system closes the gap between theory and practice making it a useful tool for developers and learners alike in the domain of blockchain and cybersecurity.

II. LITERATURE SURVEY

Hyla et. al [1] studied blockchain timestamping for digital document authentication. Their study emphasized the importance of preserving digital signatures in the long term without the necessity of repeated re-timestamping. They presented blockchain-based solutions to provide a tamper-proof and verifiable solution, contrasting permissioned and permissionless blockchain models in terms of security and efficiency. Saleh et. al [2] designed a blockchain framework for authenticating educational certificates. They pointed out the growing problem of fake academic credentials and the necessity of an open, decentralized verification system. Their framework leverages the Hyperledger Fabric blockchain, with a focus on authentication, authorization, confidentiality, privacy, and ownership to facilitate secure certificate issuance and verification.

Aziz et. al [3] investigated the use of blockchain cryptography to secure digital banking transaction records. They used RSA encryption and SHA-256 hashing to increase the security of transactions and avert cyber-attacks. They tested the accuracy and integrity of transaction data using mathematical permutation techniques and concluded that key pair cryptography improved the reliability of digital banking systems significantly. Fang et. al [4] presented an extended review of digital signature schemes under blockchain technology focusing on their responsibility for information non-repudiation. The authors classified various digital signature techniques, such as aggregate, group, ring, blind, and proxy signatures, and discussed their security properties as well as their performance. Their study aimed to guide the development of more efficient and secure digital signature algorithms for blockchain applications.

Anusha et. al [5] suggested an improved digital signature scheme for IoT and blockchain use cases and presented the Optimized Extreme Performance Edwards Curve Accumulated Signature scheme. Their work compares existing signature schemes such as the Elliptic Curve Digital Signature Algorithm (ECDSA) against the suggested scheme and presents improvements in signing and verification times, transaction rate, and block verification efficiency. The work emphasizes the need for security and efficiency in blockchain-based IoT systems. Shingari et. al [6] investigated the role of digital signatures in ensuring data integrity via RSA encryption and

SHA-1 hashing. They had based their study on the use of cryptographic techniques in ensuring security to digital communication by preventing tampering with data and illegal modifications. The research highlighted message authentication and non-repudiation as essential aspects in secure data exchange.

Gupta et. al [7] investigated blockchain-based online document verification to secure and avoid fraud. They suggested a decentralized solution based on Ethereum-based blockchain networks, hashing methods, and digital signatures to authenticate and verify the integrity of digital documents. Their research proved that blockchain can effectively reduce risks involved in document forgery and delayed verification. Choudrie et. al [8] provided varied research contributions that focused on novel ICT applications. Some of the prominent studies given were the description of blockchain usage in secure digital transactions and data privacy. The conference offered research on AI, cybersecurity, and blockchain applications with a focus on more usage of these technologies in intelligent systems.

Prakash et. al [9] suggested an effective implementation of PKI-based digital signatures with RSA and a host of hash functions (MD5 and SHA versions). It is worth noting that their work points out the significance of digital signatures for identifying unauthorized data alteration and verifying the identity of the signatory in cryptographic protocols. They explored the architecture of Public Key Infrastructure (PKI) and how it facilitates the secure nature of transactions by citing RSA along with any chosen hash function performance. Through their research, they demonstrated how their developed program benefited from an extremely increased time when signing as well as when verifying. Wang et. al [10] proposed a blockchain-based social network privacy data protection framework. Their research centered on the growing threats of user data leakage in social networks and suggested a model based on blockchain timestamps, hash functions, asymmetric encryption, and digital signatures. They classified user privacy data and built various protection schemes to prevent privacy threats. Their results indicated that although blockchain improves data security and immutability, it also has high computational overheads, which need to be tackled by future studies.

Agarwal et. al [11] co-edited a collection of studies in artificial intelligence for intelligent healthcare applications. The book discusses different AI-based innovations in healthcare, such as machine learning-based predictive modeling, intelligent healthcare applications, and security issues in AI-based medical systems. Their work highlights the convergence of AI, IoT, and blockchain to enhance healthcare decision-making, diagnosis, and data privacy. The research gives us a glimpse of new trends in AI-based healthcare and the pitfalls of applying such technologies. Adabala et al. [12] explained combining visual cryptography with AES and RSA encryption for secure digital images. Their work focused on visual secret sharing's property of decrypting through human vision, and measured the performance of the system in Peak Signal-to-Noise Ratio (PSNR) and encryption/decryption time. Results showed that

Vijjapu et al. [13] designed Medicare, a secure web application to hold medical records with multi-level security. It used role-based access control, RSA-encrypted chat, digital signatures, and secure authentication to preserve records and prevent misutilization. MongoDB stored files encrypted, giving a practical means to enhance security in e-healthcare systems. Chethana et al. [14] analyzed password security storage with a comparison of hash functions like SHA-1, MD-5, Salted MD-5, SHA-256, and SHA-512. The study proved that SHA algorithms were superior to MD-5 and Salted MD-5, and SHA-512 and SHA-256 had distinct advantages. Four new hashing functions were also suggested by the combination of other hashing functions, providing additional strength to password security and making password cracking difficult.

III. PROPOSED METHODOLOGY

A. Overview of System Architecture

The proposed work is a blockchain-based safe cryptocurrency transaction system with a modular architecture. It contains a front-end built in React and Tailwind CSS rolling on a PostgreSQL database fit for use with Supabase. This secure cryptocurrency transaction service promotes confidentiality, integrity of the transaction, and authentic identity of the user with a multistage encryption model and blockchain validation process.

A. Overview of System Architecture

The proposed work is a blockchain-based safe cryptocurrency transaction system with a modular architecture. It contains a front-end built in React and Tailwind CSS rolling on a PostgreSQL database fit for use with Supabase. This secure cryptocurrency transaction service promotes confidentiality, integrity of the transaction, and authentic identity of the user with a multistage encryption model and blockchain validation process.

Fig. 1. System Architecture

The system structure can be represented by an n-layered stack diagram:

The wallet module enables the creation of a personal public-private key pair by the user. The public key is used as the user's blockchain address, while the private key is saved securely and used to digitally sign. The wallet displays:

Fig. 2. Dashboard UI

- **Public Key:** The user's own unique blockchain address.
- **Balance:** Presented in CRYPTU units, retrieved according to unspent UTXOs.
- **Fund Transfer Form:** Accepts a recipient public key and transfer amount.

Fig. 3. Wallet Section UI

After the recipient key is pasted and the amount for transfer is entered, a layered encryption and signing process is started to protect data confidentiality and integrity before submission.

D. Encryption and Digital Signature Workflow

The core security of the system is its encryption engine, whose four-step processing of transaction information is as follows:

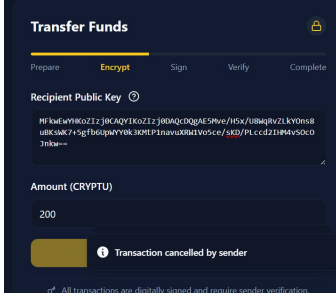


Fig. 4. Transaction Encryption UI

- 1) **Vigenère Cipher:** Utilized for carrying out a polyalphabetic substitution on the plaintext public key using a fundamental educational-level of encryption.
- 2) **AES (Advanced Encryption Standard):** Utilized to perform symmetric block cipher encryption of the message with an effective shared secret key for privacy.
- 3) **RSA Encryption:** Performs asymmetric encryption of the transaction metadata in such a way that only the intended recipient (with the appropriate private key) can decrypt.
- 4) **MD5 Hashing:** The output is MD5 hashed to ensure the data fingerprint may be matched later for verification. While MD5 is deprecated for use in critical operations, it is being used here for instructional purposes of illustrating hashing methods.

This encrypted and signed transaction is then forwarded to the transaction engine

E. Digital Signature Verification

Prior to committing a transaction, the system asks the user to digitally sign the message. This Signing:

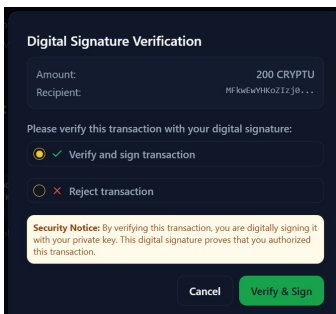


Fig. 5. Digital Signature Prompt

- Verifies the sender's identity.
- Associates the transaction with their private key.
- Guarantees non-repudiation, that is, the user cannot deny sending the transaction later.

Supabase checks the signature when it processes it on the back end. On a digital signature verification dialog, a user will be prompted to confirm and sign, using their stored private key.

F. UTXO Model for Transaction Validation

Each transaction spends existing UTXOs (Unspent Transactions Outputs) and creates a new one. The system prevents double-spending by:

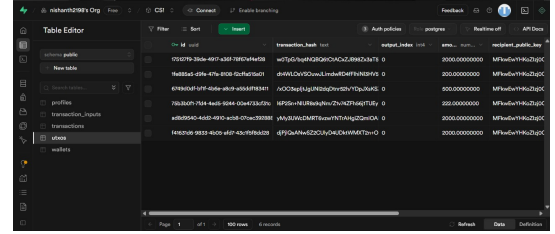


Fig. 6. Supabase — UTXO

- Limiting users to spending only their unspent outputs.
- Marking UTXOs as "spent" when they are used.
- Making a new UTXO entry for the recipient for the exact transferred amount.

Supabase includes a UTXO table that records output indexes, values, and public keys of recipients allowing for traceability and financial integrity across all actions.

G. Supabase Database Schema

The ongoing data of the platform is managed with Supabase in the following important tables:

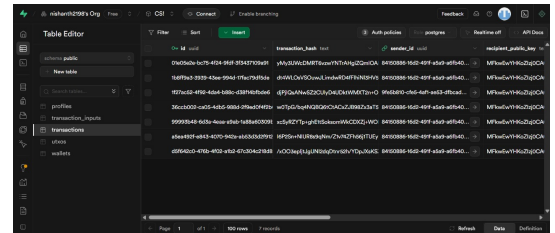


Fig. 7. Supabase — Transactions Table

- **wallets:** contains public keys, user identifiers, and current balances.
- **transactions:** logs all transfer activity with transaction_hash, amount, status (pending, confirmed, rejected), and digital signatures.
- **utxos:** maps all unspent transaction outputs for wallet public keys.
- **transaction_inputs:** (optional module) could be extended to maintain inputs tracking UTXO consumption.
- **profiles:** maintains user information and session data.

The screenshot shows the Tableau Desktop interface. On the left is a sidebar with navigation icons and a 'Table Editor' tab. The main workspace displays a table of data. The table has the following columns: 'Filter', 'Auth package', 'permissions', 'Realtime of', and 'API Docs'. The data rows include a header row with 'public_key' and 'secret_key' columns, and several rows of data with various alphanumeric strings.

Filter	Auth package	permissions	Realtime of	API Docs
	public_key		secret_key	
Relat0N-1005-4579-45d-0300036927	9740162-04d-45d-0300036927	9740162-04d-45d-0300036927	9740162-04d-45d-0300036927	9740162-04d-45d-0300036927
45d017b-45d-45d-0300036927	45d017b-45d-45d-0300036927	45d017b-45d-45d-0300036927	45d017b-45d-45d-0300036927	45d017b-45d-45d-0300036927
5af45e4e-3305-45d-0300036927	5af45e4e-3305-45d-0300036927	5af45e4e-3305-45d-0300036927	5af45e4e-3305-45d-0300036927	5af45e4e-3305-45d-0300036927

Fig. 8. Supabase — Wallets Table

Supabase supports real time and safe backend storage to make transaction progression and monitoring wallet balances completely transparent.

H. Encryption Lab (Educational Module)

An additional feature, the Encryption Laboratory, demonstrates how cryptographic algorithms work in real time. It includes interactive explanations and sample operations for:

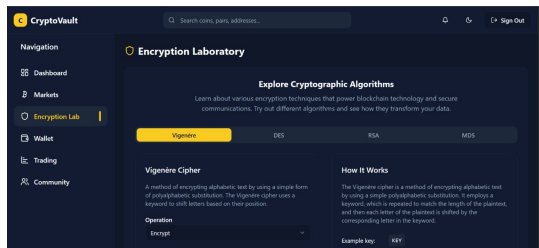


Fig. 9. Encryption Laboratory UI

- Vigenère Cipher
- DES
- RSA
- MD5

Each method includes explanation cards, example inputs, and real-time encryption/decryption demonstration. This reinforces the learning goals of the system.

I. Real-Time Status Monitoring

Each transaction goes through a state-based lifecycle controlled through Supabase:

The screenshot shows the NebulaGraph UI's 'Table Editor' for the 'public' database. The table 'table' is displayed with 10 rows of data. The columns are: id, name, age, sex, height, weight, income, education, occupation, and status. The data is as follows:

id	name	age	sex	height	weight	income	education	occupation	status
01000001	John Doe	35	Male	175	70	50000	High School	Software Engineer	Confirmed
01000002	Jane Smith	28	Female	160	55	60000	Bachelor's	Marketing Manager	Pending
01000003	Mike Johnson	42	Male	180	85	40000	High School	Construction Worker	Confirmed
01000004	Sarah Lee	30	Female	165	60	55000	Bachelor's	Teacher	Pending
01000005	David Kim	38	Male	170	75	45000	High School	Sales Representative	Confirmed
01000006	Emily White	25	Female	155	50	65000	Bachelor's	Data Analyst	Pending
01000007	Robert Brown	45	Male	185	90	35000	High School	Truck Driver	Confirmed
01000008	Lisa Green	32	Female	162	58	52000	Bachelor's	Human Resources	Pending
01000009	James Black	40	Male	178	80	42000	High School	Warehouse Worker	Confirmed
01000010	Alice Davis	27	Female	158	52	62000	Bachelor's	Product Designer	Pending

Fig. 10. Supabase — Status View

- **Pending:** Signed but unconfirmed transaction.
- **Confirmed:** Verified, signed, and added to the blockchain ledger.
- **Rejected:** Signature or balance problems; transaction was rejected.

These states are represented both in frontend UI and backend Supabase views, allowing complete traceability and trust.

IV. RESULTS

A. Visualization of Encryption in Fund Transfer Process

The execution of the cryptocurrency transfer is shown in real-time, in Fig. 11, illustrating the infrastructure’s hardening during a transaction by providing encryption frameworks for achieving datastream secrecy inside the CryptoVault application. After the user inputs the recipient’s public key and transfer amount, it is passed through multiple encryption layers in order to send the transfer.

In the graphic, the “Encrypt” tab is currently active in the transfer pipeline, indicating separation between the logical stages as follows: *Prepare* → *Encrypt* → *Sign* → *Verify* → *Complete*. The recipient’s public key is shown as an encoded Base64 string that has been encrypted using all four layering mechanisms. For instance, this field shows that the key was converted into bits using:

- Vigenère Cipher for first polyalphabetic encryption.
- AES to perform secure symmetric encryption.
- RSA to permit only the recipient user to decrypt the message.
- MD5 hashing as a method for checking message integrity.

At the bottom, a warning of “Transaction cancelled by sender” appears. This using cancel transaction indicates the system’s ability to terminate and negates transactions that are ‘manually cancelled’ by the users or serially refused from completing the verification. In this way, the user can maintain end-to-end control of the security pipeline, even when the transaction is interrupted while the system is still performing up to and intermediate encryption stage.

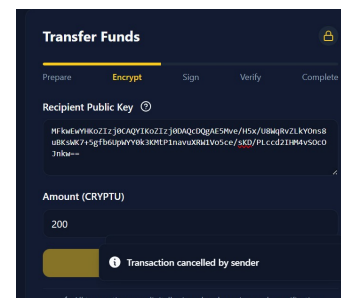


Fig. 11. Transfer Funds UI showing the encryption phase of transaction processing.

B. Data Flow Diagram of Secure Transaction Layer

Fig. 12 provides a visual summary of the encryption-decryption pipeline as implemented in the backend of CryptoVault. Once a transaction is initiated from the UI, it follows a structured sequence of transformation steps.

First, the recipient public key and transaction amount are passed through the Vigenère encryption function for symbolic obfuscation. Next, the AES algorithm applies symmetric

encryption with a fixed-length key derived from the session context. RSA encryption follows to provide confidentiality with the public key of the receiver. Finally, the encrypted output is hashed using MD5, producing a digest used to verify data integrity during storage and transmission.

This multistage encryption stack is triggered entirely on the client side and is secured with digital signatures before it is accepted by the Supabase backend.

The screenshot shows the Mahout 0.12.3 GUI with the Table Editor for the 'table' table. The table has 10 columns: id, src, dest, src_weight, dest_weight, signature, type, status, and app_desc. The status column shows various states like 'confirmed', 'pending', and 'rejected'.

Fig. 12. Backend data flow for multi-layer encryption and secure submission.

C. Output Status Handling

The platform provides feedback for every transaction state: success, pending, confirmed, rejected, or cancelled. As shown in Figure 11, the UI updates in real-time to reflect the result of each step. If a sender cancels the operation mid-way or fails digital verification, the system halts the pipeline and logs the outcome with a status flag in the database. This secure handling mechanism ensures that only valid, signed, and encrypted transactions are finalized.

V. CONCLUSION AND FUTURE WORK

A. Conclusion

In this work, we have described a secure cryptocurrency transaction validation application based on blockchain principles, developed in JavaScript (React), TypeScript, and Supabase. The application leverages a simplified model of blockchain, combined with multi-layered encryption, digital signatures and a UTXO for cryptocurrency transaction verification procedure. It represents both the computational strength of blockchain technology combined with educational merit, when coupled with both classical and current cryptographic algorithms.

The application integrally represents secure fund transfers using a pipeline involving four different stages of cryptography, namely Vigenère Cipher, AES, RSA, and MD5. The overall message integrity, authentication and non-reputation of the four combined technologies ensures secure communication. We created a simple interface for users, where they can manage wallets, sign transactions, check their live balances and log transactions using Supabase's security stacked back-end storage. The overall implementation of each transaction includes, in prescribed order of life-cycle, the transaction (to be) validated, encrypted, digitally signed, confirmed or rejected against the previously prescribed rules and behaviours of agreed /disagreed actions.

The added educational functionality of the application, a Encryption Laboratory allows users to explore different encryption options and how both methods can be valuable to secure communications. By enabling users to be presented with blockchain concepts and crypto currency in real time with 'live' user interactions, CryptoVault is a live prototype and a valuable learning tool for students and researchers as well enthusiasm users of blockchain.

B. Future Work

While CryptoVault provides a comprehensive demonstration of blockchain-backed encryption and transaction validation, several enhancements are envisioned for future development:

- **Smart Contract Integration:** Incorporating programmable smart contracts to automate conditional transactions and build decentralized applications (dApps) directly on the platform.
- **Asynchronous Blockchain Consensus:** Developing a more advanced consensus algorithm (e.g., Proof-of-Stake or Proof-of-Authority) to replace the current simplified block validation process.
- **Cross-Chain Interoperability:** Extending the system to support interoperability with other blockchain networks like Ethereum, Solana, or Bitcoin using blockchain bridges.
- **Encrypted Chat and Identity Verification:** Adding decentralized identity (DID) frameworks and secure peer-to-peer encrypted messaging modules to enable private communications alongside transactions.
- **Tokenomics and NFTs:** Introducing a native utility token and support for non-fungible tokens (NFTs) to simulate trading and asset issuance in a sandbox environment.
- **Mobile Application Support:** Developing a cross-platform mobile app to make secure blockchain operations accessible via smartphones.

Through continued refinement and feature expansion, CryptoVault aims to evolve into a fully functional educational blockchain suite that demonstrates the convergence of cryptographic theory and decentralized systems in a real-world context.

VI. REFERENCES

- [1] T. Hyla and J. Pejaš, “Long-term verification of signatures based on a blockchain,” *Computers & Electrical Engineering*, vol. 81, p. 106523, 2020.
- [2] O. S. Saleh, O. Ghazali, and M. E. Rana, “Blockchain based framework for educational certificates verification,” *Journal of Critical Reviews*, 2020.
- [3] N. Aziz, R. Rodiah, and H. Susanto, “Encrypting of digital banking transaction records: a blockchain cryptography security approach,” *International Journal of Computer Applications*, vol. 174, no. 24, pp. 21–26, 2021.
- [4] W. Fang, W. Chen, W. Zhang, J. Pei, W. Gao, and G. Wang, “Digital signature scheme for information non-repudiation in blockchain: a state of the art review,” *EURASIP Journal on Wireless Communications and Networking*, vol. 2020, pp. 1–15, 2020.
- [5] R. Anusha and R. Saravanan, “Revolutionizing signature scheme: the enhanced Edward ElGamal extreme performance accumulate signature approach for IoT and blockchain applications,” *Soft Computing*, pp. 1–24, 2025.

- [6] I. Shingari and S. S. Verma, "Achieving data integrity by forming the digital signature using RSA and SHA-1 algorithm," *IOSR Journal of Computer Engineering*, 2013.
- [7] A. Gupta, C. Khobragade, M. Gaidhane, and C. Pawar, "Online document verification using blockchain," *International Journal of Creative Research Thoughts*, vol. 11, no. 6, Jun. 2023.
- [8] J. Choudrie, P. N. Mahalle, T. Perumal, and A. Joshi, "ICT for intelligent systems," *Proceedings of ICTIS*, vol. 1, 2024.
- [9] S. Prakash and M. Purohit, "An efficient implementation of PKI architecture based digital signature using RSA and various hash functions (MD5 and SHA variants)," *IOSR Journal of Computer Engineering*, vol. 15, pp. 27–33, 2013.
- [10] P. Wang, J. Zhu, and Q. Ma, "Private data protection in social networks based on blockchain," *International Journal of Advanced Networking and Applications*, vol. 14, no. 4, pp. 5549–5555, 2023.
- [11] P. Agarwal, K. Khanna, A. A. Elngar, A. J. Obaid, and Z. Polkowski, eds., "Artificial Intelligence for Smart Healthcare," in *EAI/Springer Innovations in Communication and Computing*, 2023, doi: 10.1007/978-3-031-23602-0.
- [12] P. A. Adabala, R. S. Venkata Gowtham, T. Jaishitha, V. S. Rishisri, and K. C. R., "Enhancing Digital Image Security using Visual Cryptography Integrated with AES and RSA Encryption Techniques," *2024 3rd Edition of IEEE Delhi Section Flagship Conference (DELCON)*, New Delhi, India, 2024, pp. 1–5, doi: 10.1109/DELCON64804.2024.10866627.
- [13] A. Vijjapu, A. P. Alagawadi, A. Maddi, and C. R. Kavitha, "Enhancing Medical Record Security: Using Role-based Access Control, Digital Signatures, and RSA Encryption," *2024 8th International Conference on Electronics, Communication and Aerospace Technology (ICECA)*, Coimbatore, India, 2024, pp. 484–488, doi: 10.1109/ICECA63461.2024.10800884.
- [14] S. Chethana, S. S. Charan, V. Srihitha, D. Radha, and C. R. Kavitha, "Comparative Analysis of Password Storage Security using Double Secure Hash Algorithm," *2022 IEEE North Karnataka Subsection Flagship International Conference (NKCon)*, Vijaypur, India, 2022, pp. 1–5, doi: 10.1109/NKCon56289.2022.10127057.
- [15] A. George, F. Ashik, A. Jose, and C. R. Kavitha, "TripleACrypt: Cryptography and Steganography for the Preservation of Ancient Discoveries," *2024 8th International Conference on Computational System and Information Technology for Sustainable Solutions (CSITSS)*, Bengaluru, India, 2024, pp. 1–6, doi: 10.1109/CSITSS64042.2024.10816889.